HOA Owner Auto Pay (Payload)

QA Test Guide & Implementation Overview

Product: PropMV + Innago-inspired owner portal
Gateway: Payload (automatic payments on mirrored invoices)
Audience: QA / UAT and software engineers
Engineering: roadmap-autopay.md + existing-architecture.md

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1. What Auto Pay is (and is not)

```
Auto Pay is | Auto Pay is not
Automatic charge on each PropMV invoice due date for enrolled owners | A separate billing schedule in Payload
One Payload mirror invoice per open PropMV invoice (balance > 0) | One combined charge that pays all open invoices at once (v1)
Card or ACH via owner's saved Payload payment method | Apple Pay / Google Pay for autopay (v1)
Confirmed by Payload webhooks -> ledger + invoice status | "Paid" when the owner toggles Auto Pay on
```

Source of truth: PropMV creates invoices, due dates, balances, and frequency. Payload only executes the charge on the mirror's due date using the customer's default payment method.

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2. End-to-end flow (reference)

```
--- CODE ---
Owner enables Auto Pay (confirm checkbox + select card/ACH)
  -> OwnerPayloadAutopay saved; Payload PM set default_payment_method=true
  -> Open PropMV invoices sync to Payload (external_uid = PropMV invoice _id)

Board/owner changes balance or due date
  -> Event-driven mirror sync updates Payload line items / due_date

On due date (Payload)
  -> Automatic charge (trigger=automatic)
  -> Webhook automatic_payment (+ authorized/processed/decline as for manual pay)
  -> PayloadPayin created if missing -> same settle rules as manual Payload pay
  -> Invoice paymentTransactions + ledger updated
  -> Mirror invoice closed after successful autopay settle
---
```

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3. Environment prerequisites

3.1 Backend (`propmv/.env`)

```
Variable | Required value for Auto Pay QA
`PAYLOAD_ENABLED` | `true`
`PAYLOAD_OWNER_AUTOPAY_ENABLED` | `true`
`PAYLOAD_API_KEY` | Sandbox or staging key
`PAYLOAD_BASE_URL` | Matching Payload environment
```

Restart the API after changes.

3.2 Frontend (`innago-inspired-refresh/.env.local`)

```
Variable | Required
`VITE_PAYLOAD_ENABLED` | `true`
`VITE_PAYLOAD_OWNER_AUTOPAY_ENABLED` | `true`
```

Primary payment provider must be Payload (not Rainforest/Stripe for owner pay path).

3.3 Payload dashboard (ops)

Register webhook URL (e.g. POST /api/webhooks/payload) with at least:

- automatic_payment
- payment, processed, authorized, decline, reject, void, refund (same as manual pay)

3.4 Test data

- HOA with active Payload processing account for owner units
- Owner user linked to unit(s) on invoice tenants
- At least one saved card or ACH in Settings -> Payment methods
- At least one open invoice with balanceDue > 0 and a due date (use sandbox due date today/tomorrow for charge tests)

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4. Where to test in the UI

```
Area | Path | What to verify
Dashboard Auto Pay card | Owner dashboard (HOA context selected) | Toggle, next charge, last attempt, change method
Settings Auto Pay | Settings -> Auto Pay | Same panel; enrollment copy
Invoices | Owner invoices list / pay flow | Manual pay still works; no blank screen on invoice section
Payment methods | Settings -> Cards / banks | Add card/ACH; display correct for autopay
```

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5. API smoke tests (optional)

Base: /api/v1/payments/payload (authenticated owner session + CSRF on mutations).

```
Method | Endpoint | Notes
GET | `/autopay?hoaId={24-char}` | Returns `enabled`, `paymentMethod`, `nextAutopay`, `lastAutopayAttempt`
POST | `/autopay/enable` | Body: `{ hoaId, paymentMethodLocalId, confirmAutopay: true }`
POST | `/autopay/disable` | Body: `{ hoaId }`
PUT | `/autopay/payment-method` | Body: `{ hoaId, paymentMethodLocalId }`
```

Enable without confirmAutopay: true -> expect 400.

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6. Functional test cases

6.1 Enrollment

```
ID | Steps | Expected
AP-01 | Open Auto Pay with flags off | Card hidden or setup message (amber) explaining env flags
AP-02 | Enable without payment method | Redirect/message to add card or bank
AP-03 | Enable with method; check confirmation box; Enable | Success toast; toggle on; method shown (card vs bank + last4)
AP-04 | Enable without checking confirmation | Button disabled; cannot submit
AP-05 | Disable Auto Pay | Toggle off; Payload default cleared; open mirrors closed in Payload (ops can verify)
```

6.2 Next charge display

```
ID | Steps | Expected
AP-10 | Enrolled + open invoice with earliest due date | Next Auto Pay shows date, invoice number, balance + fee + total charge
AP-11 | No open balance | No next block or null from API
AP-12 | Two open invoices | Next date = earliest due date (not sum of both in one charge)
```

6.3 Mirror sync (balance / due date)

```
ID | Steps | Expected
AP-20 | Enable autopay with open invoice | PropMV invoice has `payloadAutopayMirror.payloadInvoiceId`; Payload invoice `external_uid` = PropMV id
AP-21 | Board reduces invoice amount or partial manual pay | Mirror updated (event sync); Payload `amount_due` <= PropMV balance + fee (ops GET invoice)
AP-22 | Invoice paid in full manually before due date | Mirror closed; no autopay charge on due date
AP-23 | Change due date on invoice | Payload mirror `due_date` updated
```

6.4 Automatic charge - card

```
ID | Steps | Expected
AP-30 | Due date reached; sandbox approves card autopay | Webhook received; invoice -> Processing then Paid when authorized/processing settles (same as manual card)
AP-31 | Ledger | Payment recorded once; balanceDue updated
AP-32 | Owner UI last attempt | Succeeded (after webhook processed)
```

6.5 Automatic charge - ACH

```
ID | Steps | Expected
AP-40 | ACH autopay initiated | Invoice Processing - not Paid until `processed` / SUCCEEDED
```

```
AP-41 | ACH completes | Paid + ledger
AP-42 | ACH return after success | Fail path; invoice not left incorrectly Paid
```

6.6 Decline / retry

```
ID | Steps | Expected
AP-50 | Declined card on due date | Invoice unpaid; last attempt Failed; Auto Pay stays enabled during Payload retry window (~5 days)
AP-51 | Retry succeeds (new txn id) | Single successful ledger entry; no duplicate settle for same txn id
```

6.7 Idempotency / safety

```
ID | Steps | Expected
AP-60 | Duplicate webhook for same transaction | No double payment on invoice
AP-61 | Autopay txn amount > current balanceDue | Settlement blocked (ops log); invoice not over-paid
```

6.8 Regression - manual pay

```
ID | Steps | Expected
AP-70 | Owner pays invoice via Pay dialog (Payload) | Unchanged flow; invoice section loads
AP-71 | Board records check/cash | Autopay mirror sync still correct
```

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7. Phase-0 sandbox checklist (required before production sign-off)

Complete in Payload sandbox with real webhooks:

1. Mirror line-item update after partial payment (amount_due matches PropMV).
2. Fee-inclusive mirror amount matches manual validate/execute for same invoice + method type.
3. Card: not Paid until authorized/processing webhook.
4. ACH: Processing until processed; return handling.
5. Decline + retry (new transaction id).
6. Duplicate automatic_payment webhook.
7. Manual full pay before due date -> no over-charge.
8. Unsafe mirror (amount_due too high) -> autopay_allowed disabled on mirror after sync guard.

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8. Data to inspect when debugging

```
Store | Field / collection | Meaning
MongoDB | `OwnerPayloadAutopay` | `enabled`, `payloadCustomerId`, PM ids per owner+HOA
MongoDB | `Invoice.payloadAutopayMirror` | `payloadInvoiceId`, `syncStatus`, `lastSyncError`
MongoDB | `PayloadPayin` | `metadata.trigger: 'automatic'`, `paymentAttemptId: automatic:{txnId}`
MongoDB | `PayloadWebhookEvent` | Idempotent webhook inbox
Payload API | `GET /invoices/{id}` | `amount_due`, `due_date`, `autopay_allowed`
Payload API | `GET /transactions/?trigger=automatic` | Autopay attempts
```

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9. Known limitations (v1)

- One charge per PropMV invoice on its own due date (not FIFO sweep across all open invoices in one autopay).
- Wallet methods not eligible for autopay enrollment.
- Production readiness depends on Phase-0 sandbox pass + webhook registration in each environment.

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10. Sign-off template

```
Environment | Date | Tester | Phase-0 (§7) | UI (§6) | Manual pay regression (AP-70-71) | Approved
Sandbox |  |  | [ ] | [ ] | [ ] | [ ]
Staging |  |  | [ ] | [ ] | [ ] | [ ]
```

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11. For software engineers - is `roadmap-autopay.md` enough?

Yes - with one companion file.

```
Document | Role
`propmv/docs/roadmap-autopay.md`   |   Primary   design   +   implementation   guide:   architecture   (§3),   enrollment,
PropMV-as-source-of-truth, webhook/idempotency (§17.2), Card vs ACH settle (§17.3), next Autopa
`propmv/docs/existing-architecture.md` | How manual Payload pay, webhooks, ledger, and Autopay hooks fit the rest of the
payments platform
This file (`qa-autopay-payload.md`) | QA / UAT procedures only - not a substitute for the roadmap
```

Recommended engineer reading order

1. roadmap-autopay.md §1-§3 (what Autopay is / end-to-end diagram)
2. roadmap-autopay.md §17 (authoritative behavioral rules)
3. roadmap-autopay.md §18 (file index -> open those modules)
4. existing-architecture.md § Owner autopay + webhook sections
5. Trace one path in code: enable -> mirror sync -> automatic_payment webhook -> settle

An engineer who follows that path can understand Autopay implementation in the codebase without additional design docs.

Document version aligns with Point 2 (§17) implementation in roadmap-autopay.md.